Delivery & Returns Analytics

Python Programming Data Analyst Professional Certificate | Capstone Project

Option B: Logistics, Delivery & Returns Insights

Business question: How well are we delivering, and what causes returns?

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GitHub repository: https://github.com/david-861/E-commerce-Delivery-and-Returns-Analytics-Python-Streamlit

Tools: Python (pandas, matplotlib, seaborn, SciPy), Plotly, Streamlit, Jupyter Notebook, Git and GitHub

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1. Project Overview

1.1 Objective, Scope, and Business Context

Objective: Answer the business question "How well are we delivering, and what causes returns?" with cleaned data, delivery and returns metrics, statistical tests, and an interactive dashboard that a manager can use without Python.

Business context: A retail e-commerce store ships about 15,000 orders through five couriers (2GO, GrabExpress, J&T, LBC, and NinjaVan) and three shipping methods (Express, Standard, and Pickup). It needs to know how fast and how reliably parcels arrive, whether any courier or shipping method performs better, and what drives customer returns. Delivery speed and reliability shape what the store can promise at checkout. Returns add handling and refund work.

Approach: I profiled all seven files before changing anything. Then I cleaned them in a fixed order and recorded every fix in a cleaning log. Rows with conflicts were flagged with True/False columns instead of deleted. This kept every original value as evidence. I built a shipment-level fact table, computed the delivery and returns metrics, exported five charts, and tested every courier and shipping method difference with a statistical test and an effect size. The results are shown in a two-page Streamlit dashboard.

Lead time was measured by its average and its spread. The spread shows how far delivery times vary from parcel to parcel. A delivery promise has to cover that spread.

Scope:

- Seven related CSV files: customers, products, orders, order items, payments, shipments, and returns.
- Ship dates from January 1, 2024 to January 31, 2026.
- Lead time covers the 11,122 shipments with status Delivered and a delivered_date.
- The return rate uses returns.csv (1,200 returns across 15,000 orders).

Deliverables: a Jupyter notebook (PC1 to PC24), seven cleaned CSVs with a 42-row cleaning log, five exported PNG charts, a two-page Streamlit dashboard (SLT1 to SLT15), a README, an executive summary, and a one-page insights summary.

1.2 Executive Summary

- 15,000 shipments: 77.9% delivered, 18.0% in transit, 3.1% returned to sender, and 1.1% lost.
- Delivered parcels take 4.0 days on average (the median is also 4 days). Delivery times are spread evenly from 1 to 7 days after shipping: about 14% of parcels arrive in 1 day, about 14% in 2 days, and so on up to 7 days. No delivery time is more common than the others.
- About 28.4% of parcels arrive within 2 days, and another 28.4% take 6 to 7 days.
- No courier is faster (ANOVA p = 0.841). Averages range from 3.97 days (NinjaVan) to 4.03 days (GrabExpress). The difference is about 1.3 hours.
- Express is not faster than Standard: 3.98 days vs 4.01 days (ANOVA p = 0.661). Both methods charge about the same shipping fee.

- Every courier is equally predictable. The standard deviation per courier is 1.97 to 2.03 days (Levene p = 0.182).
- Every courier loses parcels and sends parcels back to the store at the same rate (p = 0.706, Cramér's V = 0.014).
- The return rate is 8.0% (1,200 of 15,000 orders).
- The courier return rate difference is real but very weak (p = 0.016, Cramér's V = 0.028). Rates range from 7.0% (2GO) to 8.8% (NinjaVan).
- Return reasons are spread almost evenly. Each reason holds 13.3% (Damaged item) to 15.2% (Changed mind) of returns.
- Delivery-related reasons make up 27.4% of returns. Product and order reasons make up 72.6%.
- "Late delivery" returns came from parcels that arrived in the same median 4 days as other returns (Mann-Whitney p = 0.97).
- Data quality: 387 exact duplicate rows were removed, and 15 flag columns mark rows with invalid values, broken keys, or business-rule conflicts.
- Top recommendations: match the Express promise to the actual 4-day delivery. Give delivery-related returns to Logistics and product or order returns to Merchandising and the warehouse. Record a promised delivery date for every order.

1.3 Notebook Screenshots

The notebook python_da_capstone_project_option_b.ipynb runs from PC1 to PC24 with Kernel > Restart Kernel and Run All Cells. The screenshots below show key outputs.



PC3: Remove Exact Duplicates

Source: Day 4, Slide 13

[3]:
```
for n in ["customers", "orders", "order_items"]:
    before = len(clean[n])
    clean[n] = clean[n].drop_duplicates()
    after = len(clean[n])
    log_fix(
        "PC3",
        n,
        f"{before - after} exact duplicate rows",
        "drop_duplicates()",
        before,
        after,
        "Exact duplicates can silently inflate counts and revenue",
        f"duplicated().sum() = {clean[n].duplicated().sum()}"
    )

for n, key in [
    ("customers", "customer_id"),
    ("orders", "order_id"),
    ("order_items", "order_item_id")
]:
    print(n , key, "| repeated IDs:", clean[n].duplicated(subset=[key]).sum())

pd.DataFrame(cleaning_log)
```

```
customers customer_id | repeated IDs: 0
orders order_id | repeated IDs: 0
order_items order_item_id | repeated IDs: 0
```

[3]:

	Code	Table	Issue	Action	Rows before	Rows after	Reason	Verified by
0	PC3	customers	50 exact duplicate rows	drop_duplicates()	5050	5000	Exact duplicates can silently inflate counts a...	duplicated().sum() = 0
1	PC3	orders	75 exact duplicate rows	drop_duplicates()	15075	15000	Exact duplicates can silently inflate counts a...	duplicated().sum() = 0
2	PC3	order_items	262 exact duplicate rows	drop_duplicates()	37763	37501	Exact duplicates can silently inflate counts a...	duplicated().sum() = 0

Figure 1. PC3: exact duplicates removed (customers 50, orders 75, order_items 262)

```
customers     (5000, 16)    duplicates: 0
products      (800, 9)      duplicates: 0
orders        (15000, 9)    duplicates: 0
order_items   (37501, 8)    duplicates: 0
payments      (15000, 6)    duplicates: 0
shipments     (15000, 12)   duplicates: 0
returns       (1200, 9)     duplicates: 0

All validation checks passed.
```

[10]:

	Table	Flag	Rows flagged	% of table
0	customers	invalid_age	48	1.0%
1	customers	invalid_email	225	4.5%
2	customers	invalid_phone	133	2.7%
3	orders	orphan_customer	43	0.3%
4	order_items	price_was_backfilled	2227	5.9%
5	order_items	invalid_order_line	372	1.0%
6	order_items	orphan_product	129	0.3%
7	shipments	undelivered_with_date	3149	21.0%
8	shipments	delivered_without_date	558	3.7%
9	shipments	shipped_before_order	7397	49.3%
10	shipments	pickup_with_courier	4924	32.8%
11	shipments	cancelled_but_delivered	938	6.3%
12	returns	refund_conflict	364	30.3%
13	returns	return_not_delivered	260	21.7%
14	returns	return_before_delivery	583	48.6%

Figure 2. PC10: validation checks passed and the flag summary table

PC11: Export Cleaned CSVs and the Cleaning Log

Source: Day 4, Slide 33

[11]:

```
for n, df in clean.items():
    df.to_csv(f"../data/cleaned/{n}_clean.csv", index = False)
log_df = pd.DataFrame(cleaning_log)
log_df.to_csv("../data/cleaned/cleaning_log.csv", index = False)

# Verify: read each file back and compare its shape with the table in memory
for n, df in clean.items():
    back = pd.read_csv(f"../data/cleaned/{n}_clean.csv")
    print(f"{n}_clean.csv {back.shape}", "------ OK" if back.shape == df.shape else "------ MISMATCH" )
print(f"\ncleaning_log.csv {log_df.shape}")
log_df
```

```
customers_clean.csv (5000, 16) ------ OK
products_clean.csv (800, 9) ------ OK
orders_clean.csv (15000, 9) ------ OK
order_items_clean.csv (37501, 8) ------ OK
payments_clean.csv (15000, 6) ------ OK
shipments_clean.csv (15000, 12) ------ OK
returns_clean.csv (1200, 9) ------ OK

cleaning_log.csv (42, 8)
```

Figure 3. PC11: seven cleaned CSVs and the cleaning log exported (42 rows)

PC22: Levene's Test for Equal Variances

Source: Not in the slides (SciPy)

Question: Does any courier or shipping method deliver with a tighter spread of lead times? (Follows PC14 and PC17.)

[22]:
```
from scipy import stats

delivered = fact[fact["in_lead_time_scope"]]
print(f"Lead time scope: {len(delivered)} Delivered shipments with a delivered_date\n")

levene_rows = []
for col in ["courier", "shipping_method"]:
    groups = [g["lead_time_days"] for _, g in delivered.groupby(col)]
    w_stat, p_value = stats.levene(*groups)
    std_dev = delivered.groupby(col)["lead_time_days"].std()
    levene_rows.append((
        col,
        len(groups),
        round(w_stat, 2),
        round(p_value, 3),
        f"{std_dev.min():.2f} to {std_dev.max():.2f}"
    ))

levene_summary = pd.DataFrame(levene_rows,
                              columns = ["group_by", "groups", "W", "p_value", "std_dev_days"])

levene_summary
```

```
Lead time scope: 11122 Delivered shipments with a delivered_date
```

[22]:

	group_by	groups	W	p_value	std_dev_days
0	courier	5	1.56	0.182	1.97 to 2.03
1	shipping_method	3	0.17	0.846	1.99 to 2.00

Figure 4. PC22: Levene's test for equal variances

PC23: One-Way ANOVA (Tukey Only If Significant)

Source: Not in the slides (SciPy)

Question: Is any courier, or Express shipping, actually faster on average? (Follows PC14 and PC13.)

[23]:
```
grand_mean = delivered["lead_time_days"].mean()
ss_total = ((delivered["lead_time_days"] - grand_mean) ** 2).sum()

anova_rows = []
for col in ["courier", "shipping_method"]:
    groups = [g["lead_time_days"] for _, g in delivered.groupby(col)]
    f_stat, p_value = stats.f_oneway(*groups)
    h_stat, kw_p = stats.kruskal(*groups)

    ss_between = sum(len(g) * (g.mean() - grand_mean) ** 2 for g in groups)
    means = delivered.groupby(col)["lead_time_days"].mean()
    anova_rows.append((
        col,
        round(f_stat, 2),
        round(p_value, 3),
        round(ss_between / ss_total * 100, 3),
        f"{means.min():.2f} to {means.max():.2f}",
        round(kw_p, 3)
    ))

    if p_value < 0.05:
        print(f"{col}: significant, Tukey test below")
        print(stats.tukey_hsd(*groups))
    else:
        print(f"{col}: p = {p_value:.3f}, not significant. No Tukey Test.")

anova_summary = pd.DataFrame(anova_rows, columns = ["group_by", "F", "p_value", "r_squared_pct", "mean_lead_time_days", "kruskal_wallis_p"])

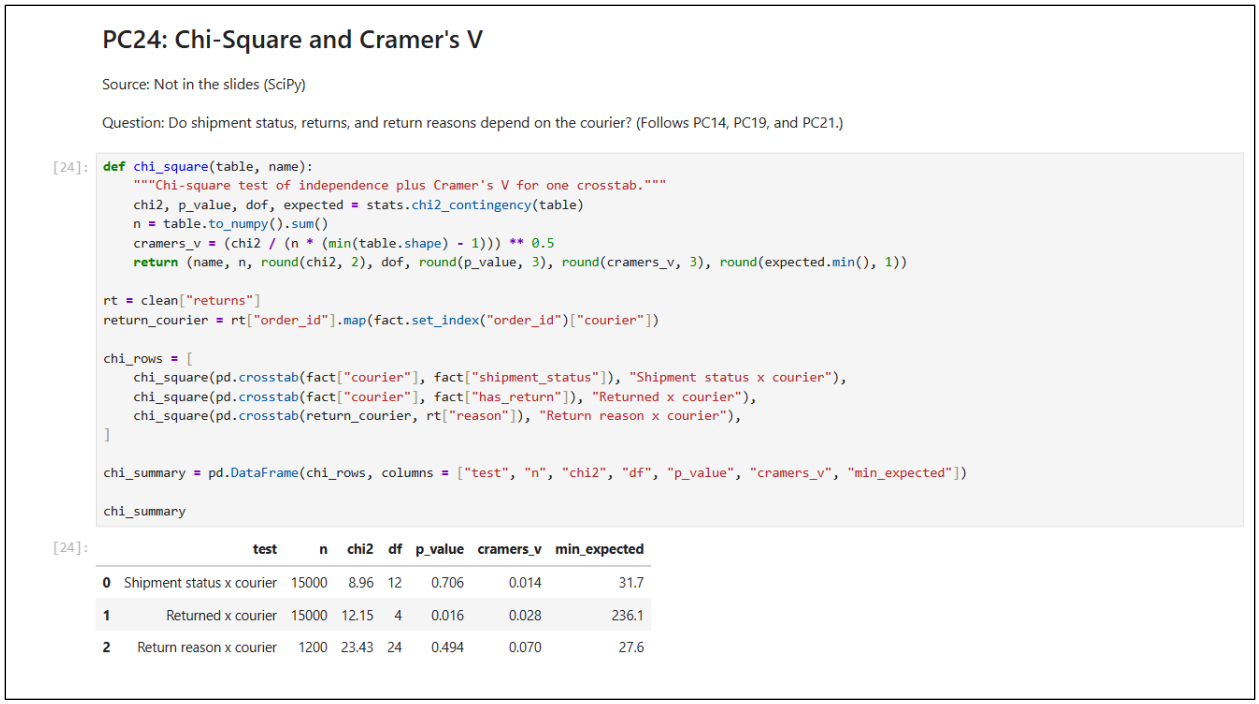
anova_summary
```

```
courier: p = 0.841, not significant. No Tukey Test.
shipping_method: p = 0.661, not significant. No Tukey Test.
```

[23]:

	group_by	F	p_value	r_squared_pct	mean_lead_time_days	kruskal_wallis_p
0	courier	0.35	0.841	0.013	3.97 to 4.03	0.842
1	shipping_method	0.41	0.661	0.007	3.98 to 4.02	0.659

Figure 5. PC23: one-way ANOVA with R^2 and the Kruskal-Wallis check



PC24: Chi-Square and Cramer's V

Source: Not in the slides (SciPy)

Question: Do shipment status, returns, and return reasons depend on the courier? (Follows PC14, PC19, and PC21.)

[24]:
```
def chi_square(table, name):
    """Chi-square test of independence plus Cramer's V for one crosstab."""
    chi2, p_value, dof, expected = stats.chi2_contingency(table)
    n = table.to_numpy().sum()
    cramers_v = (chi2 / (n * (min(table.shape) - 1))) ** 0.5
    return (name, n, round(chi2, 2), dof, round(p_value, 3), round(cramers_v, 3), round(expected.min(), 1))

rt = clean["returns"]
return_courier = rt["order_id"].map(fact.set_index("order_id")["courier"])

chi_rows = [
    chi_square(pd.crosstab(fact["courier"], fact["shipment_status"]), "Shipment status x courier"),
    chi_square(pd.crosstab(fact["courier"], fact["has_return"]), "Returned x courier"),
    chi_square(pd.crosstab(return_courier, rt["reason"]), "Return reason x courier"),
]

chi_summary = pd.DataFrame(chi_rows, columns = ["test", "n", "chi2", "df", "p_value", "cramers_v", "min_expected"])

chi_summary
```

[24]:

	test	n	chi2	df	p_value	cramers_v	min_expected
0	Shipment status x courier	15000	8.96	12	0.706	0.014	31.7
1	Returned x courier	15000	12.15	4	0.016	0.028	236.1
2	Return reason x courier	1200	23.43	24	0.494	0.070	27.6

Figure 6. PC24: chi-square tests with Cramér's V

1.4 Cleaned Data Files

All cleaned files are in data/cleaned/. Flag columns were added as True/False columns. The cleaned files have more columns than the raw files. The only rows removed were 387 exact duplicates.

File	Rows	Columns	Flag columns added
customers_clean.csv	5,000	16	invalid_age, invalid_email, invalid_phone
products_clean.csv	800	9	None
orders_clean.csv	15,000	9	orphan_customer
order_items_clean.csv	37,501	8	price_was_backfilled, invalid_order_line, orphan_product
payments_clean.csv	15,000	6	None
shipments_clean.csv	15,000	12	undelivered_with_date,delivered_without_date, shipped_before_order, pickup_with_courier, cancelled_but_delivered
returns_clean.csv	1,200	9	refund_conflict, return_not_delivered, return_before_delivery
cleaning_log.csv	42	8	Columns: Code, Table, Issue, Action, Rows before, Rows after, Reason, Verified by

2. Dashboard Screenshot and Exported Charts

2.1 Streamlit Dashboard

The dashboard has two pages. The Executive Overview page has filters for Ship Month, Courier, and Shipping Method, five KPI cards, and six charts. The Statistical Analysis page shows the seven tests on all 15,000 shipments and ignores the filters.

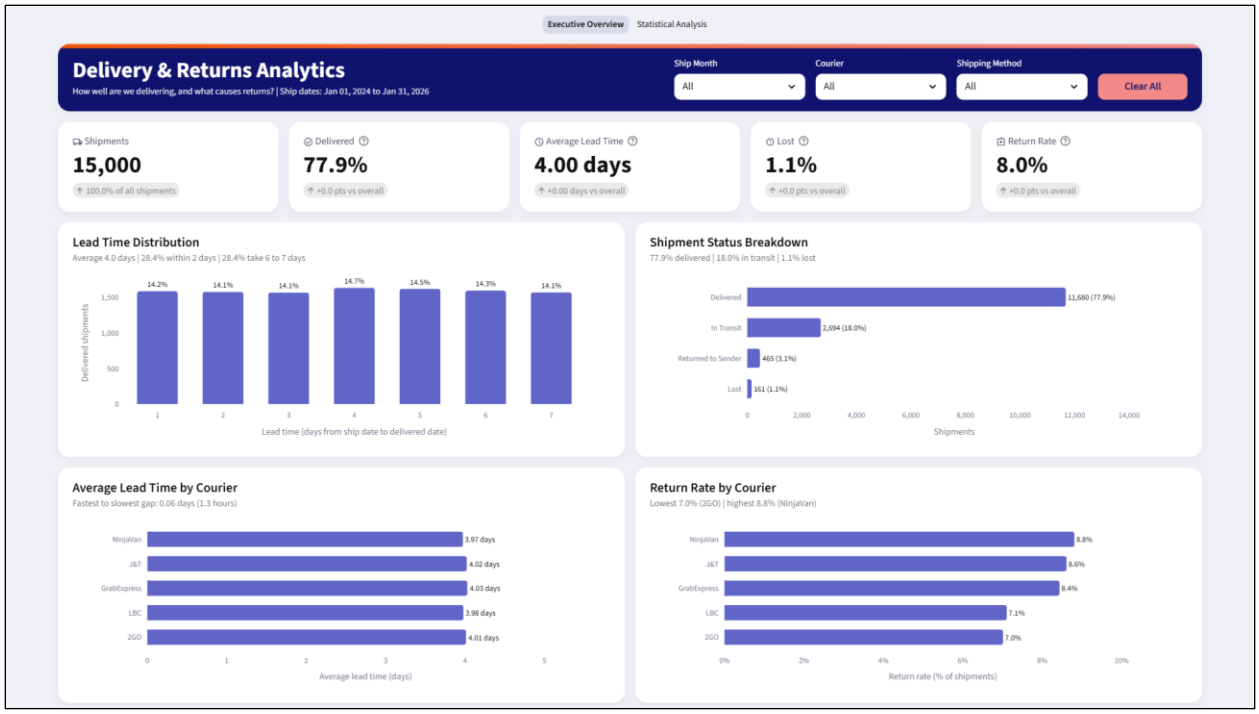


Figure 7. Executive Overview page (1920 x 1080)

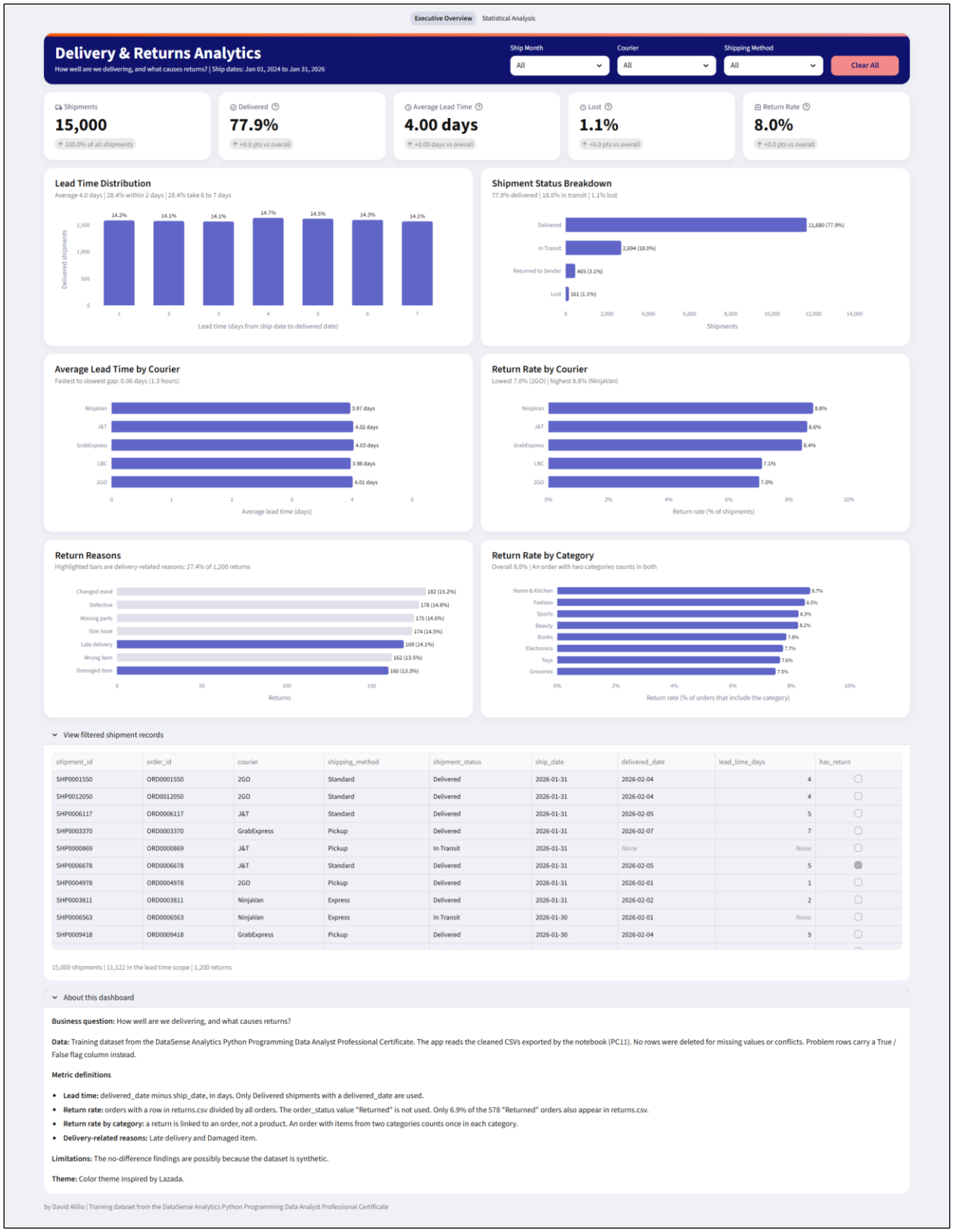


Figure 8. Executive Overview page (full page with the records table and About section open, 1920 x 2485)

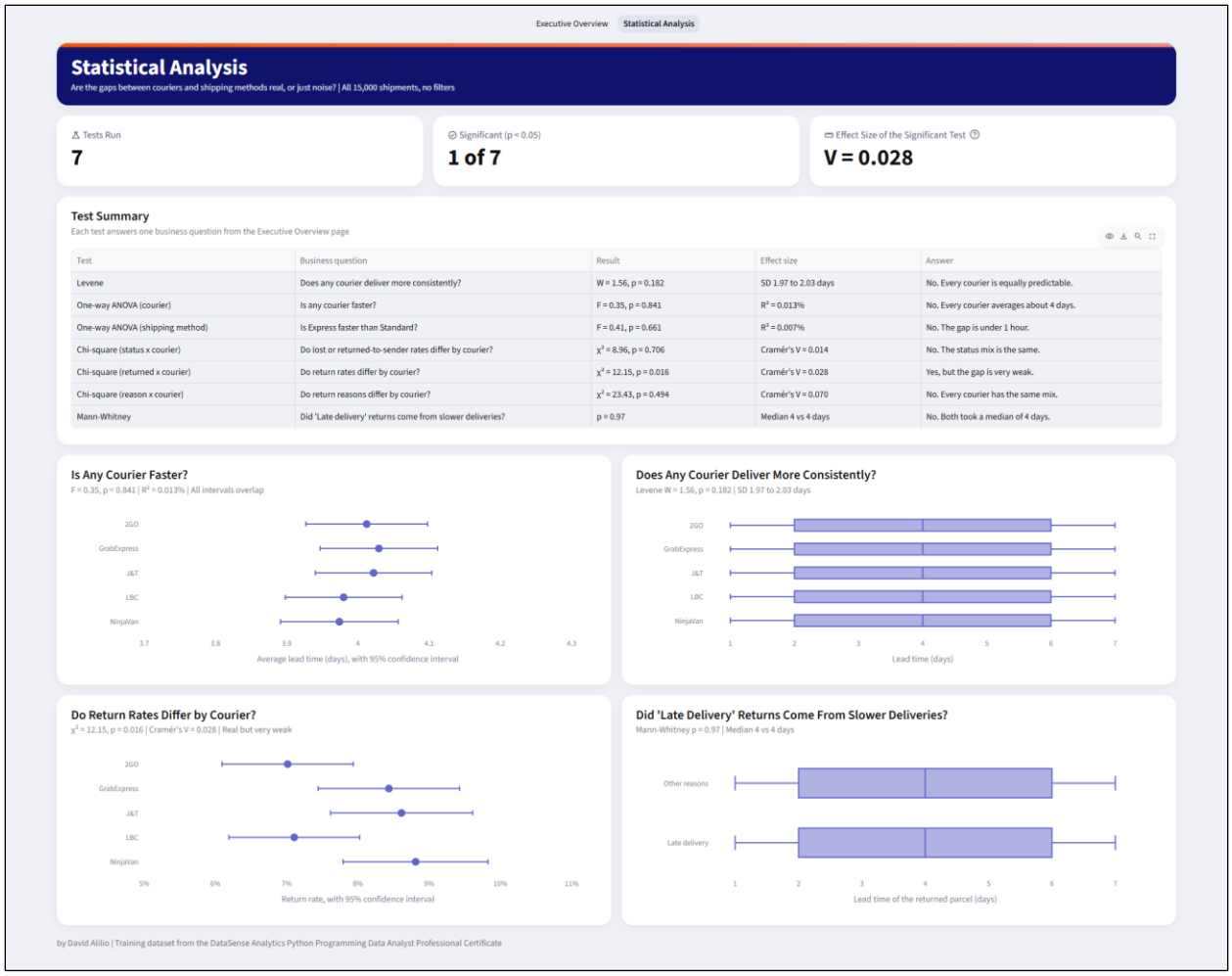


Figure 9. Statistical Analysis page (full page, 1920 x 1515)

2.2 Exported Charts (PNG)

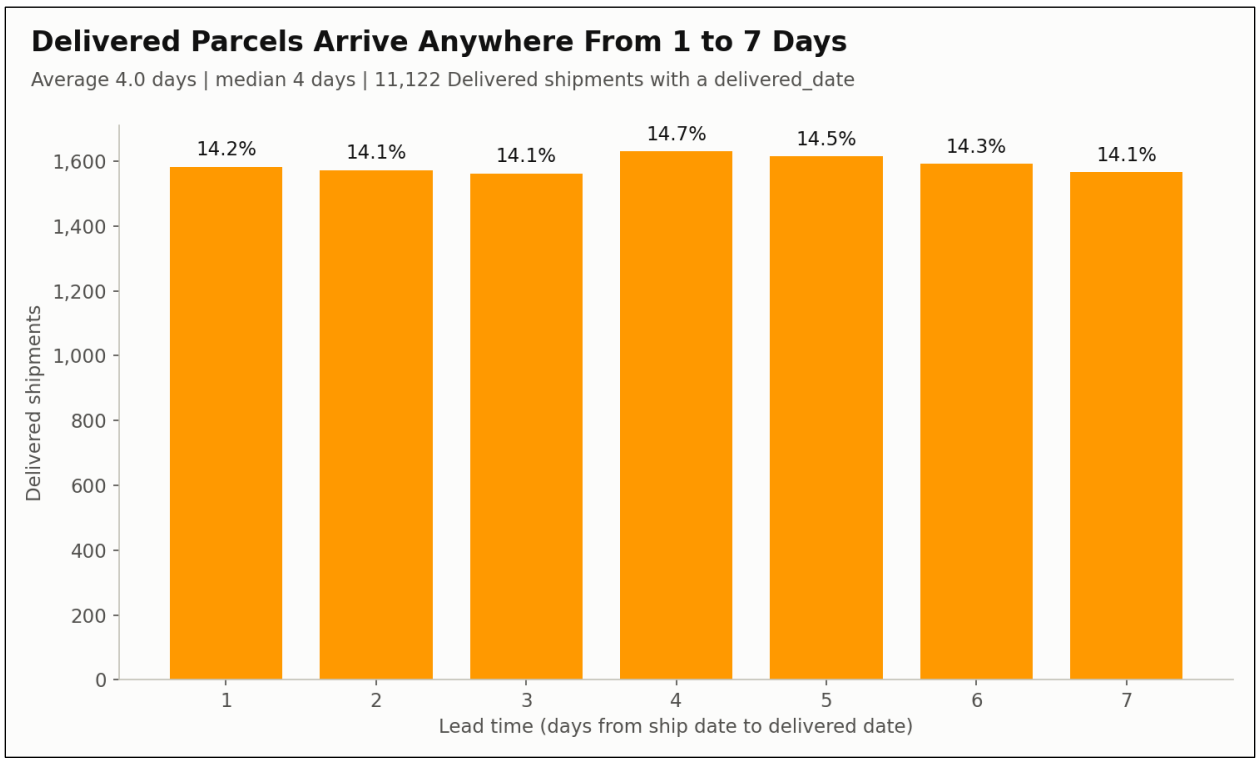


Figure 10. Lead Time Distribution (PC17)

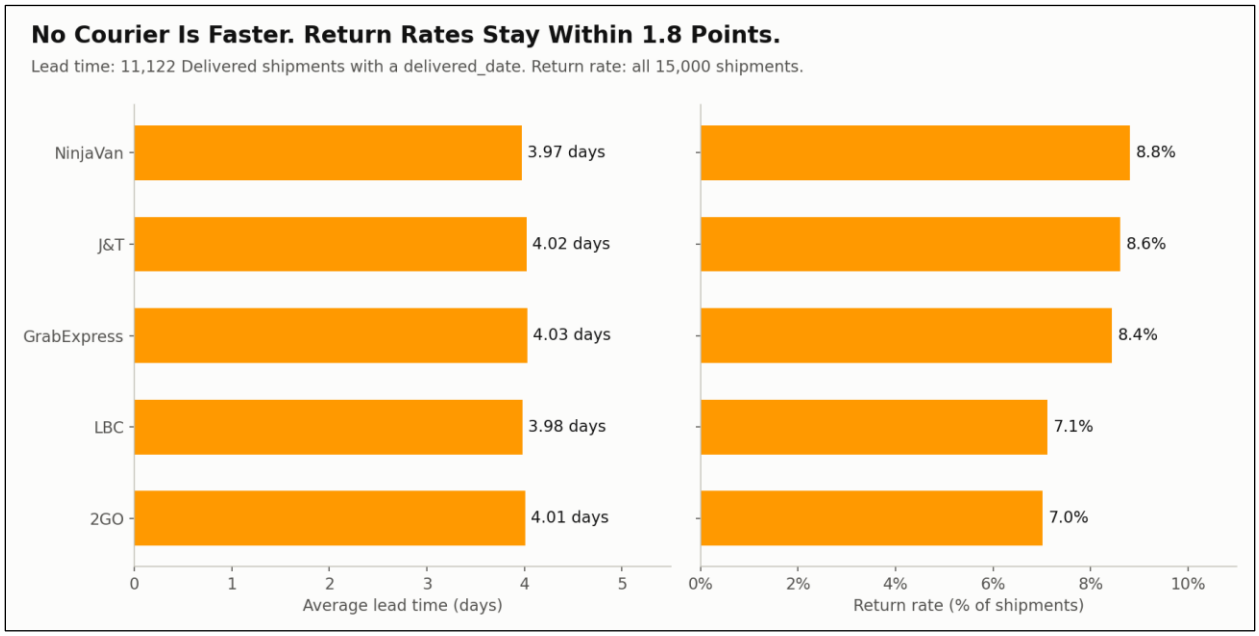


Figure 11. Courier Comparison (PC18)

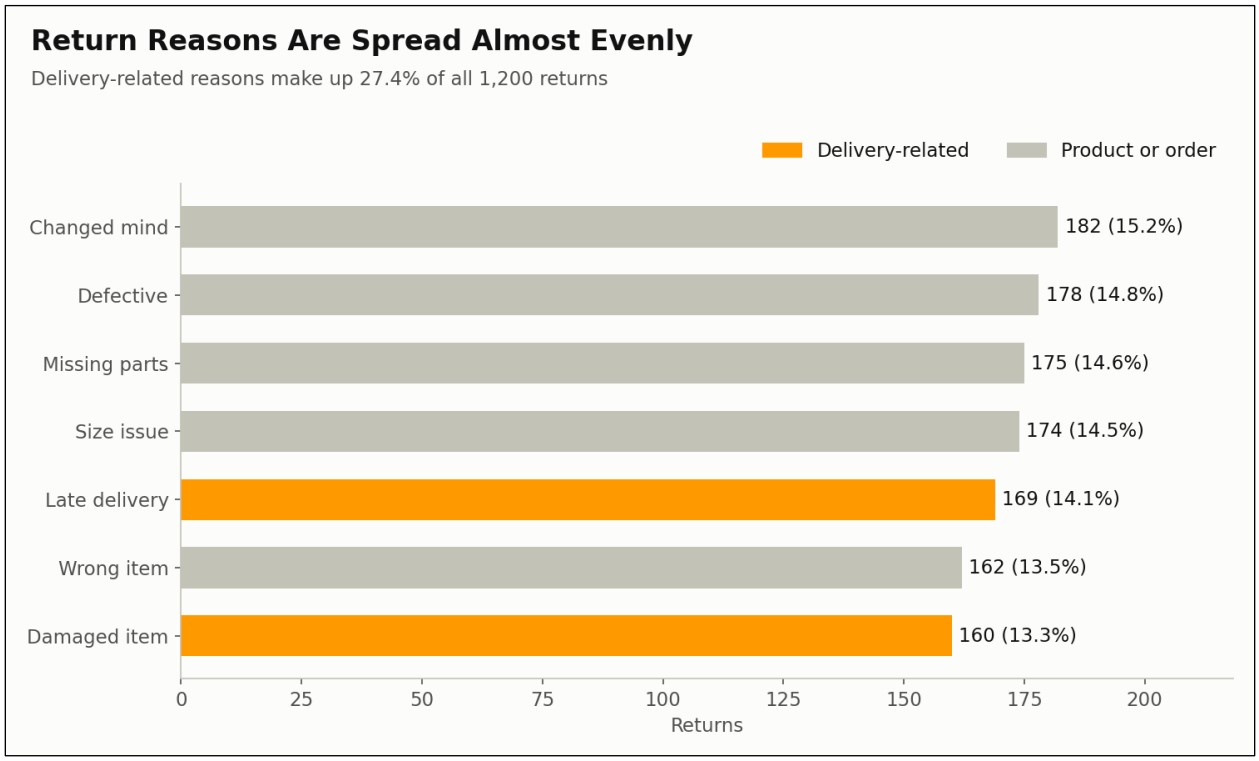


Figure 12. Return Reasons (PC19)

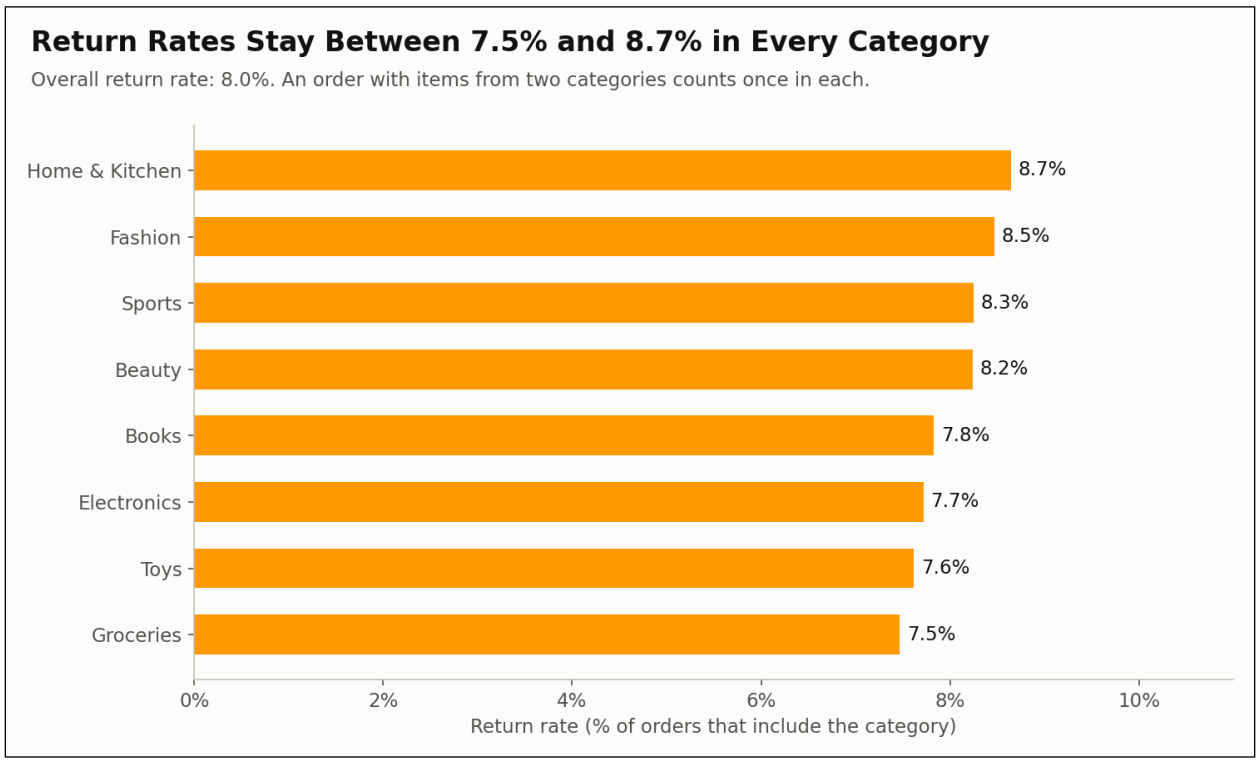


Figure 13. Return Rate by Category (PC20)

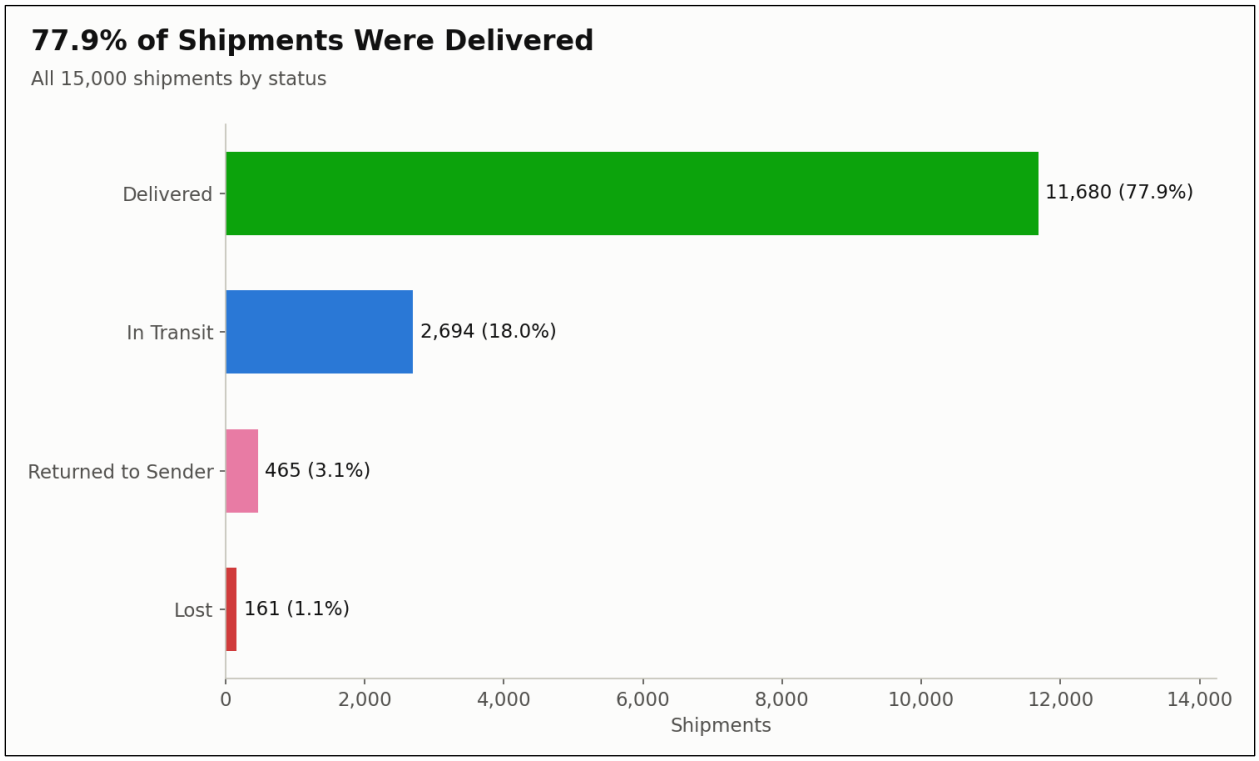


Figure 14. Shipment Status Breakdown (PC21)

3. Data Analysis and Methodology

3.1 Data Preparation and Cleaning

Every fix was written to a cleaning log with the code tag, table, issue, action, rows before, rows after, reason, and how it was verified. Two rules guided the cleaning:

- **Flag, do not delete.** Rows with invalid values or conflicts keep their original values and get a True/False flag column. Only exact duplicates were removed.
- **Missing does not mean zero.** Blanks were filled only when the right value was known. Other blanks stay blank. The isna() check already finds them.

Code Tag	Step	Result
PC1	Profile every file	Shapes, data types, blanks, duplicates, and label variants checked before any change
PC2	Working copies and cleaning log	Raw files kept untouched. Every fix is written to the log (42 rows).
PC3	Remove exact duplicates	customers 50, orders 75, order_items 262 removed. 0 repeated IDs after.
PC4	Convert currency text	Money values stored as text with ₱ and commas converted to numbers
PC5	Standardize labels	18 category labels to 8. Gender variants to Male, Female, Unknown.
PC6	Parse mixed date formats	All dates parsed with 0 failures
PC7	Handle missing values	"Unknown" for blank labels. 2,227 missing purchase prices backfilled from the product list price and flagged.
PC8	Flag invalid values and conflicts	12 flag columns for impossible values and fields that disagree within or across tables
PC9	Check cross-file key integrity	43 orders (CUST999999) and 129 order lines (PROD00000) flagged
PC10	Validate the cleaned tables	Unique IDs, numeric money columns, date columns, and label sets all passed
PC11	Export	Seven cleaned CSVs and the cleaning log, with shapes verified after reading the files back

Data conflicts found

Table	Flag	Rule	Rows	%
customers	invalid_age	Age below 0 or above 100	48	1.0%
customers	invalid_email	Email not in name@domain.ext format	225	4.5%
customers	invalid_phone	Phone number shorter than 10 digits	133	2.7%
orders	orphan_customer	customer_id not found in customers (CUST999999)	43	0.3%
order_items	price_was_backfilled	Blank unit_price_at_purchase filled with the product list price	2,227	5.9%
order_items	invalid_order_line	Quantity of 0 or less	372	1.0%
order_items	orphan_product	product_id not found in products (PROD00000)	129	0.3%
shipments	undelivered_with_date	Status is not Delivered but a delivered_date exists	3,149	21.0%
shipments	delivered_without_date	Status is Delivered but delivered_date is blank	558	3.7%
shipments	shipped_before_order	ship_date is earlier than the order date	7,397	49.3%
shipments	pickup_with_courier	Pickup order has a courier shipment	4,924	32.8%
shipments	cancelled_but_delivered	Cancelled order has a delivered shipment	938	6.3%
returns	refund_conflict	Yes or Partial with no refund amount, or No with a refund amount	364	30.3%
returns	return_not_delivered	Return linked to a shipment that is not delivered	260	21.7%
returns	return_before_delivery	Return requested before the delivered_date	583	48.6%

Two of these flags shape the analysis. First, 49.3% of ship dates are earlier than the order date. For this reason, lead time starts at the ship date. Second, Pickup orders have courier shipments in this dataset. Pickup lead time appears in the tables. No finding is based on it.

3.2 Shipment-Level Fact Table

PC12 builds one row per shipment by merging shipments with orders. The merge uses validate="one_to_one". If a key repeats, the merge stops with an error. It never adds extra rows silently. The fact table also adds three calculated columns: in_lead_time_scope, lead_time_days, and has_return. The dashboard (SLT3) also adds ship_month for the Ship Month filter.

3.3 Key Metrics and Definitions

Metric	Definition	Value
Shipments	Rows in shipments.csv after cleaning	15,000
Delivered %	Shipments with status Delivered divided by all shipments	77.9%
Lead time	delivered_date minus ship_date (in days). Delivered shipments with a delivered_date only.	4.0 days average
Within 2 days	Percentage of delivered parcels with a lead time of 2 days or less	28.38% (3,156 parcels)
6 to 7 days	Percentage of delivered parcels with a lead time of 6 or 7 days	28.39% (3,158 parcels)
Lost %	Shipments with status Lost divided by all shipments	1.1%
Return rate	Returns in returns.csv (1,200) divided by all orders (15,000). The order_status column also has a "Returned" value. It was not used. The two sources disagree: only 40 of the 578 "Returned" orders have a row in returns.csv.	8.0%
Return rate by category	Orders that include the category and have a return divided by all orders that include the category. An order with items from two categories counts in both.	7.5% to 8.7%
Delivery-related returns	Returns with reason Late delivery or Damaged item	27.4%
p-value	The probability of a difference at least this large if there were no real difference. Below 0.05 counts as significant.	$p < 0.05$
R^2	The percentage of lead time differences explained by the group (courier or shipping method)	0.013% and 0.007%
Cramér's V	A 0-to-1 scale of how strong a chi-square relationship is	0.014 to 0.070

3.4 Dashboard Filters and Interactivity

- Filters for Ship Month, Courier, and Shipping Method recalculate every KPI card and chart. A Clear All button resets them.
- Each KPI card compares the filtered value with the overall value.
- The courier charts keep all five couriers visible and highlight the selected courier. The selected courier is always shown next to its peers.
- Each chart has a one-line takeaway that updates with the filters.
- The Statistical Analysis page uses all 15,000 shipments and ignores the filters. Every result matches the notebook.
- **Color placeholder system:** every color is defined once in a THEME dictionary (SLT2). Every chart and style rule refers to a color by its name. This works like a base DAX measure that other measures reuse. A theme change edits only the placeholders.

3.5 Statistical Techniques

Every difference between couriers or shipping methods was tested before it was called a finding. Each result is reported with an effect size. A statistically real difference can still be too small to act on.

Test	Purpose in this project
Levene's test	Checks whether the spread of lead time is equal. It also decides which ANOVA to use. Equal spreads allow a standard one-way ANOVA. Unequal spreads would need Welch's ANOVA.
One-way ANOVA with R^2	Compares average lead time across couriers and shipping methods. R^2 shows how much of the lead time difference the group explains. Tukey's test runs only when the ANOVA is significant.
Kruskal-Wallis	A backup test that does not need bell-shaped data. It checks both ANOVA results.
Chi-square with Cramér's V	Compares status, return, and return reason percentages across couriers. Cramér's V shows how strong the relationship is.
Mann-Whitney	Compares the lead time of "Late delivery" returns with the lead time of other returns

3.6 Statistical Test Results

Test	Business question	Result	Effect size	Answer
Levene	Does any courier deliver more consistently?	$W = 1.56$, $p = 0.182$	SD 1.97 to 2.03 days	No. Every courier is equally predictable.
One-way ANOVA (courier)	Is any courier faster?	$F = 0.35$, $p = 0.841$	$R^2 = 0.013\%$	No. Every courier averages about 4 days.
One-way ANOVA (shipping method)	Is Express faster than Standard?	$F = 0.41$, $p = 0.661$	$R^2 = 0.007\%$	No. The 50-minute difference is not significant.
Chi-square (status x courier)	Do lost or returned-to-sender rates differ by courier?	$\chi^2 = 8.96$, $p = 0.706$	Cramér's V = 0.014	No. Every courier has about the same lost and returned-to-sender rates.
Chi-square (returned x courier)	Do return rates differ by courier?	$\chi^2 = 12.15$, $p = 0.016$	Cramér's V = 0.028	Yes, but the difference is very weak.
Chi-square (reason x courier)	Do return reasons differ by courier?	$\chi^2 = 23.43$, $p = 0.494$	Cramér's V = 0.070	No. Every courier has the same mix of return reasons.
Mann-Whitney	Did "Late delivery" returns come from slower deliveries?	$p = 0.97$	Median 4 vs 4 days	No. "Late delivery" returns and other returns both took a median of 4 days.

Kruskal-Wallis agrees with both ANOVAs ($p = 0.842$ for courier, $p = 0.659$ for shipping method). The smallest expected count in the chi-square tests is 27.6. A common rule of thumb asks for more than 5 per cell.

4. Key Takeaways

- **Delivery time is flat.** Lead time is spread evenly from 1 to 7 days. About 14% of parcels take 1 day, about 14% take 2 days, and so on up to 7 days. No delivery time is more common than the others. The store can only promise a wide 1 to 7 day window today.
- **Couriers and shipping methods perform the same.** Average lead time and spread show no meaningful difference across the five couriers or the three shipping methods. Lost and returned-to-sender rates are also the same across the five couriers.
- **Express does not deliver faster than Standard.** Express parcels arrive in 3.98 days on average. Standard parcels arrive in 4.01 days.
- **Returns are mostly about the product or the order.** 72.6% of returns cite product or order reasons. Only 27.4% cite delivery.
- **"Late delivery" is likely about customer expectations.** Parcels returned for late delivery arrived as fast as other returned parcels.
- **The one significant result is too small to act on.** Return rates differ by courier ($p = 0.016$). Cramér's V is only 0.028.
- **The shipment dates are not reliable.** 49.3% of shipments have a ship date before the order date. 21.7% of returns are linked to shipments that were not delivered. Flagging these rows kept the analysis honest about what the data can and cannot show.

5. Insights

Each finding uses both dashboard pages. The Executive Overview shows the numbers. The Statistical Analysis page tests whether a difference is real. Insights 8 and 9 come from the Executive Overview charts. Insight 10 comes from the data cleaning flags.

Insight 1: Every Courier Is Equally Predictable

Test result	Lead time by courier: $W = 1.56$, $p = 0.182$. By shipping method: $W = 0.17$, $p = 0.846$. Standard deviation per courier: 1.97 to 2.03 days. Per shipping method: 1.99 to 2.00 days.
Insight	With each of the five couriers, about 28% of parcels arrive within 2 days and about 28% take 6 to 7 days. The same holds for Express, Standard, and Pickup. Switching couriers will not narrow the delivery window the store can promise.
Importance	A delivery promise depends on the average delivery time. It also depends on how far delivery times spread around that average. No courier and no shipping method offers a tighter spread today. Every delivered parcel arrived within 7 days of shipping.
Recommendations	**Short-Term:** Use the 1 to 7 day range as the delivery window in the one-category checkout pilot (Recommendation 3). **Mid-Term:** Pilot a 3-day delivery target with one courier. Compare its spread with the other couriers over the same period.
Potential business impact	Aligned the checkout delivery promise with what every courier already delivers.
What it does not show	Lead time starts at the ship date. It does not include the time between the order and the shipment.

Where to see it	Notebook PC14 and PC22. Executive Overview: Lead Time Distribution. Statistical Analysis: lead time spread by courier.

Insight 2: No Courier Is Faster

Test result	F = 0.35, p = 0.841, R^2 = 0.013%. Average lead time: 3.97 days (NinjaVan) to 4.03 days (GrabExpress). Kruskal-Wallis check: p = 0.842. No Tukey test because the result is not statistically significant.
Insight	All five couriers deliver in about 4 days on average. The fastest and slowest couriers differ by about 1.3 hours (80 minutes). Courier explains only 0.013% of the differences in lead time.
Importance	Speed is not a reason to prefer one courier over another. Courier choice can rest on other factors such as cost and coverage. This dataset does not contain those factors.
Recommendations	**Short-Term:** Do not switch couriers or pay more for one courier based on speed claims alone. **Mid-Term:** Collect cost per parcel and area coverage for each courier. Compare couriers on those factors next.
Potential business impact	Avoided a courier switch for a speed gain of about 1.3 hours.
What it does not show	It covers delivered parcels only. Lost and In Transit parcels have no lead time.
Where to see it	Notebook PC14 and PC23. Executive Overview: Average Lead Time by Courier. Statistical Analysis: average lead time with 95% confidence intervals.

Insight 3: Express Is Not Faster Than Standard

Test result	F = 0.41, p = 0.661, R^2 = 0.007%. Average lead time: Express 3.98 days, Standard 4.01 days, Pickup 4.02 days. Kruskal-Wallis check: p = 0.659. Levene: W = 0.17, p = 0.846.
Insight	Express parcels arrive in 3.98 days on average. Standard parcels take 4.01 days. The difference is about 50 minutes. The average shipping fee is also about the same (Express ₱85.51, Standard ₱85.92).
Importance	A customer who picks Express expects a faster parcel. In this data, that customer gets the same 4-day delivery as Standard. This mismatch is a likely trust risk.
Recommendations	**Short-Term:** Review what the Express option promises at checkout. Match the wording to the actual 4-day average. **Mid-Term:** Pilot a true Express service with one courier. Compare its lead time with Standard over the same period.
Potential business impact	Protected customer trust by matching the Express label to actual delivery times.
What it does not show	The dataset cannot show whether Express orders ship sooner after the order. 49.3% of shipments have a ship_date earlier than the order date. That step cannot be measured reliably. Pickup orders also have courier shipments in this dataset (flag pickup_with_courier). Pickup orders should not need a courier. Pickup lead time appears in the tables. No finding is based on it.

Where to see it	Notebook PC13 and PC23. Executive Overview: the Shipping Method filter with the Average Lead Time card. Statistical Analysis: test summary table.

Insight 4: Lost and Returned-to-Sender Rates Are the Same Across Couriers

Test result	$\chi^2 = 8.96$, df = 12, p = 0.706, Cramér's V = 0.014.
Insight	Overall, 77.9% of shipments were delivered, 18.0% were in transit, 3.1% were returned to sender, and 1.1% were lost. Each courier stays close to these percentages. Lost parcels range from 0.8% (NinjaVan) to 1.2% (GrabExpress and J&T).
Importance	No courier loses or returns more parcels than the others. Lost and returned-to-sender parcels are a shared problem across all couriers.
Recommendations	**Short-Term:** Review a sample of returned-to-sender parcels to find their common cause. Check for wrong addresses first. **Mid-Term:** Track lost and returned-to-sender rates per courier every month. Act only on a courier that moves away from the shared rate.
Potential business impact	Focused loss-reduction work on the shared process instead of a courier switch.
What it does not show	Why parcels were lost or returned to sender. The dataset has no reason field for these statuses.
Where to see it	Notebook PC14 and PC24. Executive Overview: Shipment Status Breakdown. Statistical Analysis: test summary table.

Insight 5: The Courier Return Rate Difference Is Real but Small

Test result	$\chi^2 = 12.15$, df = 4, p = 0.016, Cramér's V = 0.028.
Insight	The percentage of shipments with a return ranges from 7.0% (2GO, 207 of 2,951) to 8.8% (NinjaVan, 263 of 2,985). The difference is real (p = 0.016) but very weak (Cramér's V = 0.028).
Importance	A 1.8 percentage point difference is too small to justify a courier switch on returns alone. Return reasons also look the same for every courier (Insight 6). The data does not point to a courier-specific problem.
Recommendations	**Short-Term:** Keep the current couriers. Do not switch based on this return rate difference. **Mid-Term:** Rerun this test every quarter. Investigate a courier only if its return rate pulls further away from the others.
Potential business impact	Avoided a courier switch based on a 1.8 percentage point difference.
What it does not show	Whether the courier is the reason for a return. The returns data has no field that links a return to courier handling.
Where to see it	Notebook PC14 and PC24. Executive Overview: Return Rate by Courier. Statistical Analysis: return rate with 95% confidence intervals.

Insight 6: Return Reasons Look the Same for Every Courier

Test result	$\chi^2 = 23.43$, df = 24, p = 0.494, Cramér's V = 0.070. Smallest expected count: 27.6.
Insight	The seven return reasons are spread almost evenly. They range from 13.3% (Damaged item) to 15.2% (Changed mind). The mix of reasons does not differ by courier.
Importance	Delivery-related reasons (Damaged item and Late delivery) make up 27.4% of returns. No courier has a bigger share of them. The other 72.6% point to the product, the order, or the customer's choice. Reducing returns needs work on products and order accuracy as much as on delivery.
Recommendations	**Short-Term:** Report return reasons in two groups every month: delivery-related (27.4%) and product or order (72.6%). **Mid-Term:** Give each group to the team that can fix it. Logistics owns delivery-related reasons. Merchandising and the warehouse own product and order reasons.
Potential business impact	Prioritized the product and order reasons behind 72.6% of returns.
What it does not show	The deeper cause behind each return reason. For example, "Changed mind" has no follow-up detail.
Where to see it	Notebook PC19 and PC24. Executive Overview: Return Reasons (highlighted bars are delivery-related).

Insight 7: "Late Delivery" Returns Did Not Come From Slower Deliveries

Test result	Median lead time 4 days for "Late delivery" returns (n = 135) and 4 days for other reasons (n = 759). Mann-Whitney p = 0.97. Only returns linked to delivered parcels are included.
Insight	169 of 1,200 returns (14.1%) cite "Late delivery." Their parcels arrived in a median of 4 days. Parcels returned for other reasons also arrived in a median of 4 days.
Importance	"Late" likely means later than the customer expected. The dataset has no promised delivery date to confirm this.
Recommendations	**Short-Term:** Pilot a clear delivery window at checkout for one category. **Mid-Term:** Record the promised delivery date for every order.
Potential business impact	Enabled a one-category test of whether clearer delivery windows lower "Late delivery" returns.
What it does not show	What delivery time the customer expected. There is no promised-date field.
Where to see it	Notebook PC15 (Part D) and PC19. Executive Overview: Return Reasons. Statistical Analysis: lead time of "Late delivery" returns vs other returns.

Insight 8: Every Category Has About the Same Return Rate

Test result	No statistical test. Return rate by category: Home & Kitchen 8.7%, Fashion 8.5%, Sports 8.3%, Beauty 8.2%, Books 7.8%, Electronics 7.7%, Toys 7.6%, and Groceries 7.5%.
Insight	The highest and lowest categories differ by only 1.2 percentage points. Every category sits close to the overall 8.0% return rate.
Importance	Returns are not concentrated in one category. A fix for a single category will not lower the overall return rate by much. The return reasons are a better place to start (Insight 6).
Recommendations	**Short-Term:** Keep the category return rate in the monthly returns report. Watch for a category that moves away from the others. **Mid-Term:** Break down the return reasons by category. A pair such as "Size issue" in Fashion may show a problem that a single category rate hides.
Potential business impact	Avoided a category-specific fix that would not move the overall return rate.
What it does not show	Which product inside a category drives returns. A return is linked to an order. It is not linked to a product. An order with items from two categories counts in both.
Where to see it	Notebook PC15 (Part C) and PC20. Executive Overview: Return Rate by Category.

Insight 9: Delivery Time Is Spread Evenly From 1 to 7 Days

Test result	No statistical test. Lead time from 1 to 7 days: 14.1% to 14.7% of delivered parcels per day. Average 4.0 days, median 4 days. 28.38% (3,156 parcels) arrive within 2 days. 28.39% (3,158 parcels) take 6 to 7 days. 42.9% take more than 4 days.
Insight	Lead time counts the days from shipping to delivery. It is not a day of the week. About as many parcels arrive in 1 day as in 7 days. No delivery time is more common than the others. We cannot say that more parcels arrive in 4 days than in 2 days.
Importance	The 4-day average hides a wide spread. A 4-day promise would be late for 42.9% of delivered parcels. Every delivered parcel arrived within 7 days of shipping.
Recommendations	**Short-Term:** Show a delivery window of 1 to 7 days at checkout instead of a single date. **Mid-Term:** Work with the couriers to shorten the 6 to 7 day tail. Track the percentage of parcels that take 6 to 7 days every month.
Potential business impact	Set a delivery promise that every delivered parcel in this data met.
What it does not show	Why some parcels take 6 to 7 days. The dataset has no route or distance field.
Where to see it	Notebook PC13 and PC17. Executive Overview: Lead Time Distribution and the Average Lead Time card.

Insight 10: Shipment Dates Are Not Reliable

Test result	No statistical test. PC8 flags: shipped_before_order 7,397 (49.3%), undelivered_with_date 3,149 (21.0%), cancelled_but_delivered 938 (6.3%), and delivered_without_date 558 (3.7%).
Insight	Almost half of the ship dates are earlier than their order dates. A parcel cannot ship before the order exists. Status and date fields also disagree in thousands of rows.
Importance	Order-to-delivered time is the wait the customer actually feels. It cannot be measured until the dates are reliable. The 4.0-day lead time covers only the part from shipping to delivery.
Recommendations	**Short-Term:** Keep the flags in every report. Report the percentage of shipments with valid dates every month. **Mid-Term:** Record ship and delivered dates from courier scans (Recommendation 5).
Potential business impact	Enabled order-to-delivered tracking once the dates are fixed.
What it does not show	The root cause. The data cannot tell whether the dates were typed wrong, came from a time zone or date format mismatch, or were generated at random.
Where to see it	Notebook PC8 output, the data conflicts table in Section 3.1, and data/cleaned/cleaning_log.csv.

6. Summary

77.9% of shipments were delivered. Delivered parcels take 4.0 days on average within a wide 1 to 7 day window. Average lead time and spread are the same across all five couriers and all three shipping methods. Lost and returned-to-sender rates are also the same across the five couriers. Express, in particular, is not faster than Standard.

The return rate is 8.0%. Return reasons and category return rates are spread almost evenly. 72.6% of returns point to the product or the order rather than the delivery. The only statistically significant result, the courier return rate difference, is too weak to act on. "Late delivery" returns came from parcels that arrived as fast as other returns. This likely points to customer expectations rather than slow couriers.

The data also has a reliability problem. 49.3% of ship dates are earlier than their order dates. For this reason, the time from order to delivery cannot be measured yet.

7. Conclusions

1. **How well are we delivering?** Consistently, but with a wide delivery window. Most parcels arrive, but the 1 to 7 day spread means the store cannot promise a tight delivery date with any current courier or shipping method.
2. **Courier choice is not the lever.** No courier is faster, more consistent, or less likely to lose parcels. Switching couriers would not narrow the delivery window.
3. **The Express option is not delivering a faster service.** Its lead time and fee match Standard. This mismatch is a likely trust risk.

4. **What causes returns?** Mostly product and order issues (72.6%), not delivery (27.4%). Return reasons look the same for every courier. Return rates differ only slightly across couriers (7.0% to 8.8%) and categories (7.5% to 8.7%).
5. **The data needs better fields.** Reliable ship and delivered dates, a promised delivery date, courier cost and coverage, and reasons for lost or returned-to-sender parcels would allow deeper analysis. The no-difference findings may be a sign that the dataset is synthetic.

8. Recommendations

In order of priority:

#	Recommendation	Short-term action	Mid-term action	Based on
1	Match the Express promise to actual delivery	Review the Express wording at checkout and match it to the 4-day average	Pilot a true Express service with one courier and compare it with Standard over the same period	Insight 3
2	Split returns by the team that can fix them	Report delivery-related (27.4%) and product or order (72.6%) returns every month	Logistics owns delivery-related returns. Merchandising and the warehouse own product and order returns.	Insights 6 and 8
3	Record a promised delivery date for every order	Test a clear delivery window at checkout in one category (for example, "Arrives in 1 to 7 days")	Record the promised date for every order to make "Late delivery" measurable. Compare the pilot category's "Late delivery" returns with other categories.	Insights 1, 7, and 9
4	Keep the current couriers and compare them on cost and coverage next	Do not switch couriers based on speed or the small return rate difference	Collect cost per parcel and coverage per courier. Rerun the return rate test every quarter.	Insights 2, 4, and 5
5	Record shipment dates from system scans	Keep the date flags. Report the percentage of shipments with valid dates every month.	Take the ship date from the courier pickup scan and the delivered date from the proof-of-delivery scan. Add a daily data quality check. Logistics owns the fix.	Insight 10

Appendix A. Limitations

- The no-difference findings may be a sign that the dataset is synthetic. The results describe this dataset only.
- Lead time starts at the ship date. The time from order to shipment cannot be measured reliably. In 49.3% of shipments, the ship date is earlier than the order date.
- The dataset has no promised or standard delivery date, no reason field for lost or returned-to-sender parcels, and no courier cost or coverage data.

Appendix B. Tools Used

Tool	Used for
Python (pandas)	Profiling, cleaning, flag columns, key-integrity checks, merges, and aggregations
matplotlib and seaborn	The five exported charts
SciPy	Levene, one-way ANOVA, Kruskal-Wallis, chi-square, and Mann-Whitney tests
Plotly and Streamlit	The interactive two-page dashboard
Jupyter Notebook (Anaconda)	Analysis notebook
Git and GitHub	Version control and the public repository

Appendix C. Error Log

Errors I hit while building this project and how I fixed them.

Tag	Where	Problem	Fix
E1	PC8	The email pattern used [A-Z-z] instead of [A-Za-z]. It flagged 4,896 emails as invalid instead of 225.	Corrected the character range and reran the notebook.
E2	PC10	The shape printout used str(df.shape)<13 instead of the format spec :<13, and the duplicate count used duplicated.sum() without the parentheses.	Changed to {str(df.shape):<13} and duplicated().sum().
E3	PC15	A misplaced bracket in it[~it["invalid_order_line"]] & ~it["orphan_product"] tried to build a 10.4 GiB array. It raised a MemoryError and crashed the kernel.	Moved the bracket so both conditions sit inside one filter.
E4	PC15	drop_duplicates() ran before keeping only the order and category columns. Home & Kitchen counted 4,522 orders instead of 4,034.	Selected [["order_id", "category"]] first and then dropped duplicates.
E5	PC19	The format spec .1f% is not valid for percentages.	Changed it to .1%.
E6	dashboard	app.py was first created as a notebook and then renamed. The file held notebook JSON instead of Python code.	Deleted it and created a plain text file named app.py.
E7	SLT8	NameError: STATUS_COLORS was deleted from SLT2 while the status chart still used it.	Changed the chart to one color, color_discrete_sequence=[THEME["accent"]].
E8	SLT2	The header CSS was typed above <style>. It showed as plain text on the page.	Moved the lines inside the <style> block.
E9	SLT2	The header stripe CSS used commas instead of semicolons (content: "",). The browser ignored those lines, and the stripe did not show.	Replaced the commas with semicolons.

Tag	Where	Problem	Fix
E10	SLT13	UnhashableParamError: @st.cache_data could not hash the fact DataFrame passed to run_tests().	Renamed the arguments to _fact and _returns. A leading underscore tells Streamlit to skip hashing an argument.
E11	SLT15	NameError: position-"top" used a minus sign. Python read position as a variable.	Changed it to position="top".
E12	SLT15	The page links stayed on the left. My Streamlit version has no stTopNav test ID. The centering CSS matched nothing.	Inspected the nav bar in the browser DevTools, found the .rc-overflow container, and centered that instead.
E13	SLT10	In the tall screenshots, the Return Reasons legend covered the "182 (15.2%)" label of the top bar.	Removed the legend and moved the color meaning into the chart subtitle ("Highlighted bars are delivery-related reasons").